Practice 1: Likelihood language

Name: ____________________ Date: __________

Throughout this workbook, dice are fair and numbered 1–6, and coins are fair. Each object is equally likely to be drawn, and each equal-sized spinner section is equally likely. P(event) means the probability of that event.

Use impossible, unlikely, equally likely, likely, or certain. Explain your choice.

1. A standard six-sided die is rolled. The result is a number from 1 through 6.

Work space

2. A bag has 9 blue tiles and 1 red tile. Which color is more likely? How do you know?

Work space

Practice 2: Sample spaces

List every possible outcome, with no repeats.

1. A coin is tossed once.

Work space

2. A spinner has equal sections labeled 1, 2, 3, and 4. It is spun once.

Work space

Practice 3: Favorable outcomes

Find the number of total outcomes and favorable outcomes.

1. A die is rolled. What is the probability of rolling a 5?

Work space

2. A die is rolled. What is the probability of rolling an even number?

Work space

Practice 4: Probability as a fraction

Write each probability as a fraction in simplest form when possible.

1. A jar has 4 green, 3 yellow, and 1 purple marble. What is P(yellow)?

Work space

2. From the same jar, what is P(not purple)?

Work space

Practice 5: Impossible and certain events

1. A bag contains only red and white counters. What is the probability of drawing a blue counter?

Work space

2. A box contains 12 pencils, all sharpened. What is the probability of choosing a sharpened pencil?

Work space

Practice 6: Complementary events

Use P(event) + P(not event) = 1.

1. A spinner has 8 equal sections; 3 are orange. Find P(not orange).

Work space

2. The probability that a bus is early is $\frac{1}{5}$. Find the probability that it is not early.

Work space

Practice 7: Comparing probabilities

1. Bag A has 2 red tiles and 6 blue tiles. Bag B has 4 red tiles and 8 blue tiles. Which bag gives a greater chance of red? Show work.

Work space

2. Compare $\frac{3}{10}$ and $\frac{1}{4}$. Which event is more likely? Explain.

Work space

Practice 8: Fairness

1. A game uses a fair coin. One player wins on heads and the other on tails. Is this a fair way to choose between them? Explain.

Work space

2. A spinner has 3 equal red sections and 1 equal blue section. Is it fair for red and blue players? Why or why not?

Work space

Practice 9: Theoretical probability

1. A fair die is rolled 60 times. About how many 6s would you expect? Explain.

Work space

2. A bag has 5 green and 5 yellow counters. In 40 draws with replacement, about how many green draws are expected?

Work space

Practice 10: Experimental probability

Experimental probability is favorable results divided by trials.

1. A coin lands heads 17 times in 30 tosses. Find the experimental probability of heads.

Work space

2. A spinner lands on blue 9 times in 25 spins. Find the experimental probability of blue.

Work space

Practice 11: Predicting from data

1. A die is rolled 120 times. A 4 appears 18 times. Use the data to predict how many 4s may appear in 600 rolls.

Work space

2. A red result occurred 12 times in 20 trials. Predict red results in 100 similar trials.

Work space

Practice 12: Tables and probability

Use the table to answer each question.

Color	Red	Blue	Green	Yellow
Count	6	5	3	2

1. How many tiles are there? Find P(green).

Work space

2. Find P(red or blue).

Work space

Practice 13: Two-step outcomes

For two independent coin tosses, list ordered outcomes.

1. List the sample space for two tosses and find P(two heads).

Work space

2. Find P(exactly one head) for two tosses.

Work space

Practice 14: Two-color draws

Choose one counter, replace it, then choose again.

1. A bag has 2 red and 1 blue counter. List the four possible color pairs and find P(red then red).

Work space

2. For the same bag and replacement, find P(one red and one blue in either order).

Work space

Practice 15: Reading claims

1. Sam says, "A probability of $\frac{7}{4}$ is possible." Is Sam correct? Explain using the probability range.

Work space

2. Lee says, "If an event has probability $\frac{1}{2}$, it must happen exactly half the time." Evaluate the claim.

Work space

Practice 16: Showing reasoning

Write a clear conclusion with an equation or fraction.

1. A box has 5 apples and 7 oranges. What is the probability of an apple? Show numerator and denominator.

Work space

2. Explain why a probability of 0 means impossible and 1 means certain. Give one example of each.

Work space

Practice 17: Word problems

1. A prize wheel has 10 equal spaces: 4 toys, 3 books, 2 games, and 1 sticker. What is P(a book or game)?

Work space

2. In a class survey, each of 24 students chooses one favorite fruit. Of these students, 15 choose apples. What fraction choose another fruit? Simplify.

Work space

Practice 18: Design a probability model

1. Design a fair spinner for 3 players. Describe the equal sections and justify fairness.

Work space

2. Design a spinner where blue has probability $\frac{1}{4}$. Describe one possible spinner.

Work space

Practice 19: Mixed review

1. A bag has 3 red, 4 blue, and 5 green cubes. Find P(not green) and explain your count.

Work space

2. A fair die is rolled 90 times. Predict the number of odd results and state the assumption used.

Work space

Practice 20: Explain and justify

1. Which is more likely: rolling a number greater than 4 on a die, or drawing a red tile from a bag with 3 red and 7 blue tiles? Compare the fractions.

Work space

2. Write a short rule for finding theoretical probability, then apply it to P(rolling less than 3) on a fair die.

Work space